

Yash Raj Sharan

+91 94714 44635 | yashrajsharan2006@gmail.com | linkedin.com/in/yashraj10
github.com/omen18 | yash-dev-seven.vercel.app | Chennai, Tamil Nadu, India
Full Stack Developer · AI & Machine Learning Undergraduate at SRMIST

SUMMARY

Full stack developer building production-style web applications end to end — responsive React + TypeScript frontends and Node.js / Express and FastAPI backends. Comfortable across REST and WebSocket API design, MySQL and MongoDB schema modelling, authentication, concurrency-safe transactions, and deploying performance-tuned apps, with applied AI/ML integrated into real product workflows. Open source contributor with merged code in IBM's Machine Learning for Kids platform.

EDUCATION

SRM Institute of Science and Technology, Chennai — B.Tech, Computer Science & Engineering (AI & ML)

2024–2028

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, Java, SQL, Rust, Swift

Frontend: React 18/19, Next.js, Vite, HTML5, CSS3, Tailwind CSS, Three.js, GSAP, Responsive UI, REST/API integration

Backend: Node.js, Express.js, FastAPI, REST APIs, WebSockets, JWT & bcrypt auth, concurrency control, transactional integrity

Databases: MySQL (schema design, indexing, atomic transactions), MongoDB

AI/ML: PyTorch, TensorFlow, ONNX Runtime, NumPy, Pandas, Generative AI / LLM integration

Tools: Git, GitHub, Postman, Vercel, Jupyter Notebook, Google Colab, Xcode

OPEN SOURCE CONTRIBUTIONS

IBM/taxinomitis — Machine Learning for Kids · Node.js, JavaScript

[PR #539 \(Merged\)](#)

Extended the isWikimedia and getImageName utilities to support Wikimedia's thumb.wikimedia.org CDN domain alongside upload.wikimedia.org, unblocking image URLs the validator previously rejected. Replaced hardcoded-URL test assertions with checks on dynamic domain and thumbnail sizing. Merged to master after code-owner review.

refinedev/refine — TypeScript

PR open — under review

Refactored core permission helpers for cleaner semantics and early exit, and expanded unit test coverage in the core package with 9 new tests (+477 lines).

PROJECTS

CutisAI — React 18, FastAPI, PyTorch, ONNX Runtime

[GitHub](#)

Skin lesion segmentation and classification system pairing a ResUNet segmentation stage with an EfficientNet-B0 classifier, served through INT8-quantized ONNX Runtime for faster CPU inference. Delivered as a React 18 dashboard over a FastAPI inference API.

AmritKrishi 2.0 — RAG, Retrieval Pipelines, Full Stack Web

[GitHub](#)

Multilingual farming assistant for Indian smallholders, delivering crop advisory, mandi price lookups, and weather-driven sowing recommendations. Answers are grounded in government agricultural datasets through a RAG pipeline rather than free-form generation.

ShowPass — React, TypeScript, Node.js, MySQL

[GitHub](#)

Full stack event ticket booking platform with real-time seat selection, JWT + bcrypt authentication, and server-validated pricing. Implemented atomic, concurrency-safe transactions and a seat-lock service that eliminates double booking under parallel load.

MacDeck — Swift, ScreenCaptureKit, VideoToolbox, Bonjour

[GitHub](#)

macOS + iOS remote control over LAN. Hardware-encoded H.264 screen streaming (ScreenCaptureKit → VideoToolbox → AVSampleBufferDisplayLayer) with a CGEvent input back-channel. Bonjour discovery, TLS-encrypted, no servers or accounts.

Developer Portfolio — React, TypeScript, Three.js, Express.js

[GitHub](#) | [Live](#)

Built a performance-tuned portfolio platform with Three.js 3D visuals, GSAP animation choreography, and live GitHub API data. Backed by an Express.js service with a secure contact pipeline and an AI "Ask Me Anything" chat widget served through an edge function.

AWARDS & CERTIFICATIONS

Networking Devices and Initial Configuration — Cisco Networking Academy · GirlScript Summer of Code (AI Track) · Research Paper — Digital Image Processing

ADDITIONAL

Languages: Hindi, English, German